INDUCTION PRINCIPLE STARTING FROM A NONEMPTY BASE SUBSPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete, with the finite affine refinement of the square handled in full

Editorial note. The label lemma-induction-principle-enlarge names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- elementary distinguished squares of algebraic spaces
- the filtration of a quasi-compact quasi-separated space by quasi-compact opens from the decent spaces chapter
- refining a distinguished square by an affine open cover so the square hypothesis applies
- descending induction on the filtration

Source reference. Original lemma 07ST: sources/stacks-chow/tags.txt, source lines 9913–9913. Prerequisite pointer invoked at proof line(s) 1738; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at proof line(s) 1731; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1643–1643. Prerequisite pointer invoked at proof line(s) 1729; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1541–1541. Prerequisite pointer invoked at proof line(s) 1769; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $W \subset X$ be a quasi-compact open subspace. Let P be a property of quasi-compact open subspaces of X. Assume that*

(1) *P holds for W, and*
(2) *for every elementary distinguished square $(W_1 \subset W_2, f : V \to W_2)$ where such that*
   (a) *W_1, W_2 are quasi-compact open subspaces of X,*
   (b) *$W \subset W_1$,*
   (c) *V is affine, and*
   (d) *P holds for W_1,*

   then P holds for W_2.

Then P holds for X.

Proof. We can deduce this from Lemma 08GQ, but instead we will give a direct argument by explicitly redoing the proof of Lemma 08DR. We will use the filtration

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

and the morphisms $f_p : V_p \to U_p$ of Decent Spaces, Lemma 07ST. We will prove that P holds for $W_p = W \cup U_p$ by descending induction on p. This will finish the proof as $W_1 = X$. Note that P holds for $W_{n+1} = W \cap U_{n+1} = W$ by (1). Assume P holds for W_{p+1}. Observe that $W_p \setminus W_{p+1}$ (with reduced induced subspace structure) is a closed subspace of $U_p \setminus U_{p+1}$. Since $(U_{p+1} \subset U_p, f_p : V_p \to U_p)$ is an elementary distinguished square, the same is true for $(W_{p+1} \subset W_p, f_p : V_p \to W_p)$. However (2) may not apply as V_p may not be affine. However, as V_p is a quasi-compact scheme we may choose a finite affine open covering $V_p = V_{p,1} \cup \ldots \cup V_{p,m}$. Set $W_{p,0} = W_{p+1}$ and

$$W_{p,i} = W_{p+1} \cup f_p(V_{p,1} \cup \ldots \cup V_{p,i})$$

for $i = 1, \ldots, m$. These are quasi-compact open subspaces of X containing W. Then we have

$$W_{p+1} = W_{p,0} \subset W_{p,1} \subset \ldots \subset W_{p,m} = W_p$$

and the pairs

$$(W_{p,0} \subset W_{p,1}, f_p|_{V_{p,1}}), (W_{p,1} \subset W_{p,2}, f_p|_{V_{p,2}}), \ldots, (W_{p,m-1} \subset W_{p,m}, f_p|_{V_{p,m}})$$

are elementary distinguished squares by Lemma 08GN. Now (2) applies to each of these and we inductively conclude P holds for $W_{p,1}, \ldots, W_{p,m} = W_p$. □